EE5239 Homework 5

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1 Written Problems

1.1 3.3.1

$$\frac{\partial f}{\partial x} = 2(x-a)+y, \quad \frac{\partial f}{\partial y} = 2(y-b)+x$$

$$2(x-a)+y=0, \quad 2(y-b)+x=0$$

$$y=-2(x-a), \quad x=-2(y-b)$$

$$x=-2(-2(x-a)-b)$$

$$x=4(x-a)+2b$$

$$x(1-4)=4a+2b$$

$$x=\frac{4a+2b}{3}$$

$$y=\frac{4b+2a}{3}$$

Critical point:

$$(x,y)=\left(\frac{4a+2b}{3},\frac{4b+2a}{3}\right)$$

Solution if $0 \leq x \leq 1, 0 \leq y \leq 1$ in the critical point:

$$f\left(\frac{4a+2b}{3},\frac{4b+2a}{3}\right)=\frac{10a^2+26ab+13b^2}{9}$$

If the critical point is not feasible, the answer is these where x or y is set to 0 or 1 if they are >0 or <1 respectively in the critical point:

$$f(0,y)=a^2+(y-b)^2$$

$$f(1,y)=(1-a)^2+(y-b)^2+y$$

$$f(x,0)=(x-a)^2+b^2$$

$$f(x,1)=(x-a)^2+(1-b)^2+x$$

$$f(0,0) = a^2 + b^2$$
$$f(0,1) = a^2 + (1-b)^2$$
$$f(1,0) = (1-a)^2 + b^2$$
$$f(1,1) = (1-a)^2 + (1-b)^2 + 1$$

1.2 3.3.3

$$\mathcal{L} = x_N + \sum_{i=0}^{N-1} (1-u_i)x_i + \sum_{i=0}^{N-1} \lambda_i\big(x_{i+1} - x_i - wu_ix_i\big).$$

$$\mathcal{L} = x_0 \prod_{i=0}^{N-1} (1+wu_i) + \sum_{i=0}^{N-1} (1-u_i)x_i + \sum_{i=0}^{N-1} \lambda_i\big(x_{i+1} - x_i - wu_ix_i\big).$$

$$\frac{\partial \mathcal{L}}{\partial u_i} = -x_i + \lambda_i w x_i \prod_{j=i+1}^{N-1} (1+wu_j),$$
$$\frac{\partial \mathcal{L}}{\partial x_i} = (1-u_i) - \lambda_i + \lambda_{i-1}(1+wu_{i-1}),$$
$$\frac{\partial \mathcal{L}}{\partial \lambda_i} = x_{i+1} - x_i - wu_ix_i.$$

$$u_i = \begin{cases} 1 & \text{if } w > 1, \\ 0 & \text{if } w < \frac{1}{N}, \\ 1 \text{ for } i < \tilde{i},\ 0 \text{ for } i \geq \tilde{i},\ \text{if } \frac{1}{N} \leq w \leq 1, \end{cases}$$

where $\tilde{i}$ satisfies:
$$\frac{1}{\tilde{i}+1} < w \leq \frac{1}{\tilde{i}}.$$

1.3 3.4.1

y is defined as:

- If $p_i > y$: Outlets with $p_i > y$ have the highest prices. Assign $s_i^* = d_i$ to maximize revenue.
- If $p_i < y$: Outlets with $p_i < y$ contribute less to revenue, so assign $s_i^* = 0$.
- If $p_i = y$: For outlets where $p_i = y$, allocate the remaining quantity to satisfy the constraint $\sum_{i=1}^{n} s_i = Q$.

To compute s_i^*, perform the following:

1. Sort Outlets: Sort the outlets by decreasing price, such that $p_1 \geq p_2 \geq \cdots \geq p_n$.
2. Compute k: Find the largest k such that:

$$\sum_{i=1}^{k} d_i \leq Q, \quad \text{but} \sum_{i=1}^{k+1} d_i > Q.$$

3. Assign s_i^*:
   - For $i = 1, \ldots, k$: Set $s_i^* = d_i$.
   - For $i = k+1$: Set $s_{k+1}^* = Q - \sum_{i=1}^{k} d_i$.
   - For $i > k+1$: Set $s_i^* = 0$.

1.4 3.4.3

The primal and dual problems are symmetric because the constraints of the primal define the feasible region of the dual and vice versa, and the objective function of one problem is a linear combination of the constraints of the other.

1. For the first pair:

$$A'x - b \geq 0, \quad \mu \geq 0, \quad \mu_i (A'x - b)_i = 0 \quad \forall i.$$

2. For the second pair:

$$A'x - b \geq 0, \quad x \geq 0, \quad \mu \geq 0, \quad A\mu \leq c,$$

$$\mu_i (A'x - b)_i = 0 \quad \forall i, \quad x_j (A\mu - c)_j = 0 \quad \forall j.$$

1. The Lagrangian for the first pair is:

$$L(x, \mu) = c'x + \mu'(b - A'x),$$

where $\mu \geq 0$. The dual function:

$$g(\mu) = \inf_x L(x, \mu) = b'\mu, \quad \text{if } A\mu = c.$$

Maximizing $g(\mu)$ gives the dual.

2. For the second pair:

$$L(x, \mu) = c'x + \mu'(b - A'x),$$

where $\mu \geq 0$. Dual feasibility $A\mu \leq c$ arises from complementary slackness.

1.5 3.4.4

The Lagrangian for the primal problem is:

$$L(x,u,v) = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}x_{ij} + \sum_{i=1}^{m} u_i \left(\alpha_i - \sum_{j=1}^{n} x_{ij}\right) + \sum_{j=1}^{n} v_j \left(\beta_j - \sum_{i=1}^{m} x_{ij}\right),$$

where:

- u_i: Lagrange multiplier for the supply constraint $\sum_{j=1}^{n} x_{ij} = \alpha_i$,
- v_j: Lagrange multiplier for the demand constraint $\sum_{i=1}^{m} x_{ij} = \beta_j$.

Reorganizing terms:

$$L(x,u,v) = \sum_{i=1}^{m}\sum_{j=1}^{n} (a_{ij} - u_i - v_j)\, x_{ij} + \sum_{i=1}^{m} u_i\alpha_i + \sum_{j=1}^{n} v_j\beta_j.$$

The dual function is obtained by minimizing the Lagrangian with respect to $x_{ij} \geq 0$:

$$g(u,v) = \inf_{x_{ij}\geq 0} L(x,u,v).$$

To ensure the Lagrangian is bounded below:

$$a_{ij} - u_i - v_j \geq 0, \quad \forall i,j.$$

Substituting this condition, the dual function becomes:

$$g(u,v) = \sum_{i=1}^{m} u_i\alpha_i + \sum_{j=1}^{n} v_j\beta_j.$$

The dual problem is:

$$\max_{u,v} \sum_{i=1}^{m} u_i\alpha_i + \sum_{j=1}^{n} v_j\beta_j,$$

subject to:

$$a_{ij} \geq u_i + v_j, \quad \forall i,j.$$

For the primal solution x_{ij}^* to be optimal, it must satisfy the complementary slackness condition:

$$x_{ij}^* \cdot (a_{ij} - u_i^* - v_j^*) = 0, \quad \forall i,j.$$

This implies:

- If $x_{ij}^* > 0$, then:
$$u_i^* + v_j^* = a_{ij}.$$

- If $x_{ij}^* = 0$, then:
$$u_i^* + v_j^* \leq a_{ij}.$$

Let $p_j = v_j^*$ be the price that demand point j is willing to pay for one unit of material delivered to it. The condition $x_{ij}^* > 0$ implies:

$$p_j - a_{ij} = u_i^*.$$

This means that the profit (or surplus) for supply point i from shipping to demand point j is maximized when $x_{ij}^* > 0$.

For any supply point i, the maximum net profit is achieved when:

$$u_i^* = \max_{k=1,\ldots,n} \{p_k - a_{ik}\}.$$

Substituting u_i^* into the complementary slackness condition gives:

$$p_j = \max_{k=1,\ldots,n} \{p_k - a_{ik}\} + a_{ij}.$$

A feasible dual solution (u^*, v^*) satisfies:

$$a_{ij} \geq u_i^* + v_j^*, \quad \forall i, j.$$

A feasible primal solution x_{ij}^* satisfies:

$$\sum_{j=1}^{n} x_{ij} = \alpha_i, \quad \sum_{i=1}^{m} x_{ij} = \beta_j, \quad x_{ij} \geq 0.$$

- If x_{ij}^* is optimal, then u_i^* and v_j^* must satisfy the complementary slackness condition:
$$x_{ij}^* \cdot (a_{ij} - u_i^* - v_j^*) = 0.$$
- This implies that for every feasible primal solution, there exists a corresponding dual solution satisfying these conditions.
- Every dual optimal solution (u^*, v^*) corresponds to a feasible primal solution x_{ij}^*.
- The optimality condition ensures that the prices $p_j = v_j^*$ and profits $u_i^* = \max_k\{p_k - a_{ik}\}$ align with the primal solution.

2 Coding Problem

I chose the task of classifying between 1s and 5s. The model used for this problem is a soft-margin SVM. This model minimizes the margin between support vectors plus a hyperparameter C times the sum of the slack variables, which are the distances from their support vector of points that are not on the correct side of their support vector. The algorithms I chose to solve this model were gradient descent and coordinate descent. My implementation of gradient descent in each iteration first calculates the margin, the distance of each point from the current decision boundary, then identifies points for which the margin is less than 1, then assigns nonzero slack variables to these points, then calculates the gradient, which is the current decision boundary minus C times the sum of the slack variables multiplied by the outcome vector multiplied by the features matrix based on the objective function, then updates the next iteration's decision boundary by subtracting the gradient by the learning rate. The coordinate descent algorithm follows a similar process but the gradient is only calculated and and the decision boundary is only updated for one coordinate of the decision boundary at a time. The results, as shown below, are very accurate on both the training and testing sets and are extremely similar for the two models.

```
SVM (Gradient Descent) - Training Accuracy: 99.68%
SVM (Gradient Descent) - Testing Accuracy: 97.88%
SVM (Coordinate Descent) - Training Accuracy: 99.62%
SVM (Coordinate Descent) - Testing Accuracy: 97.88%
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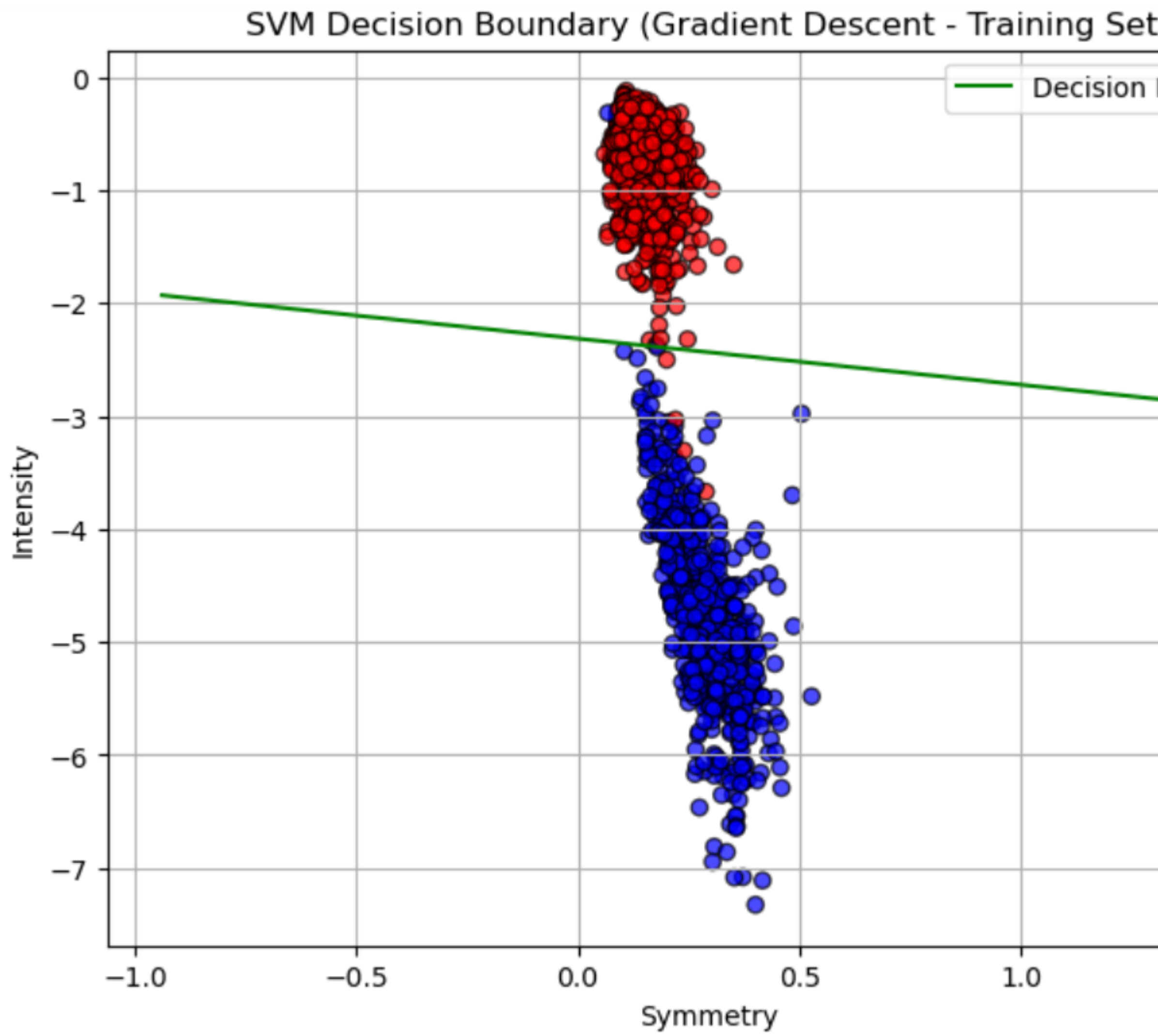
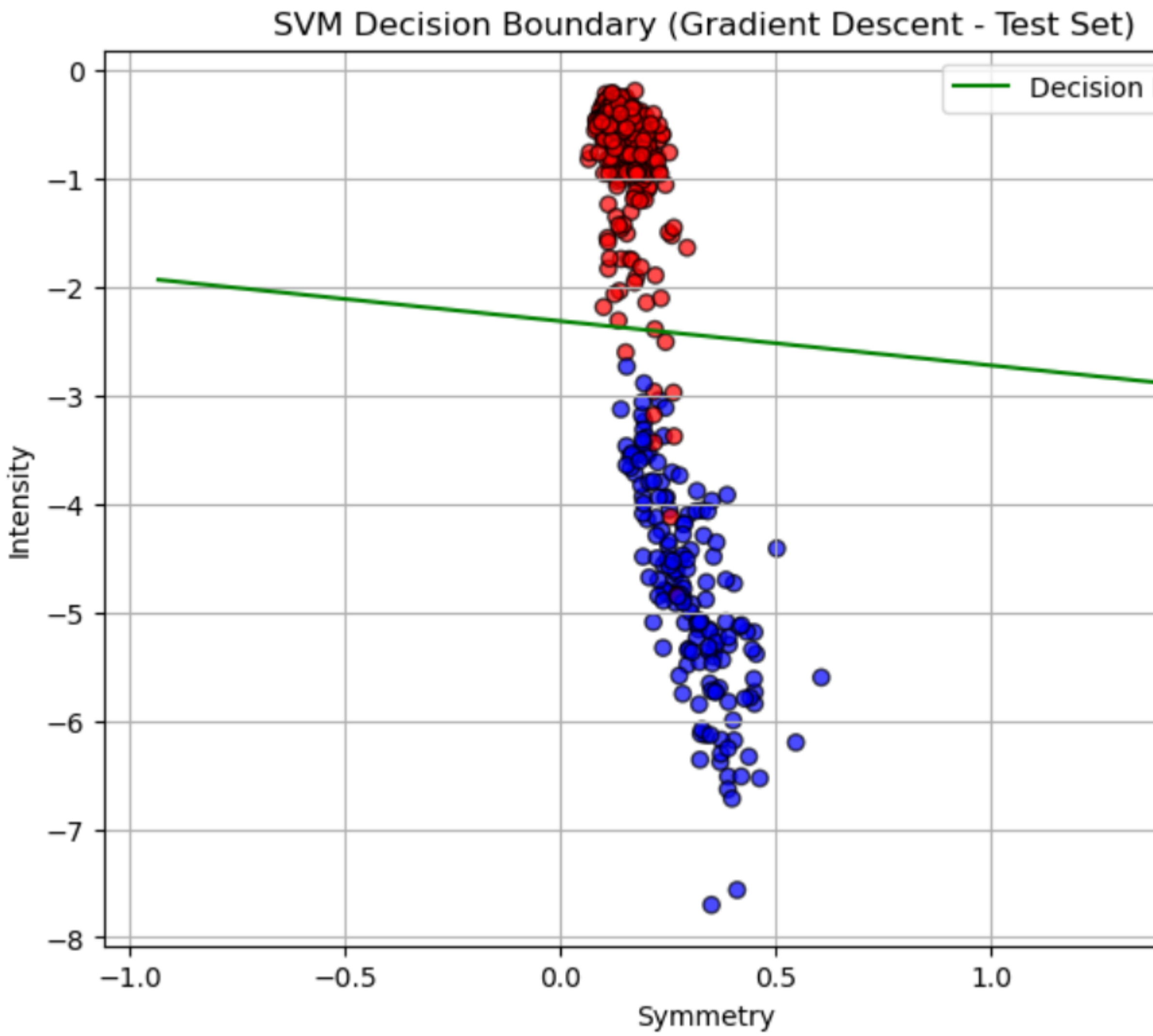
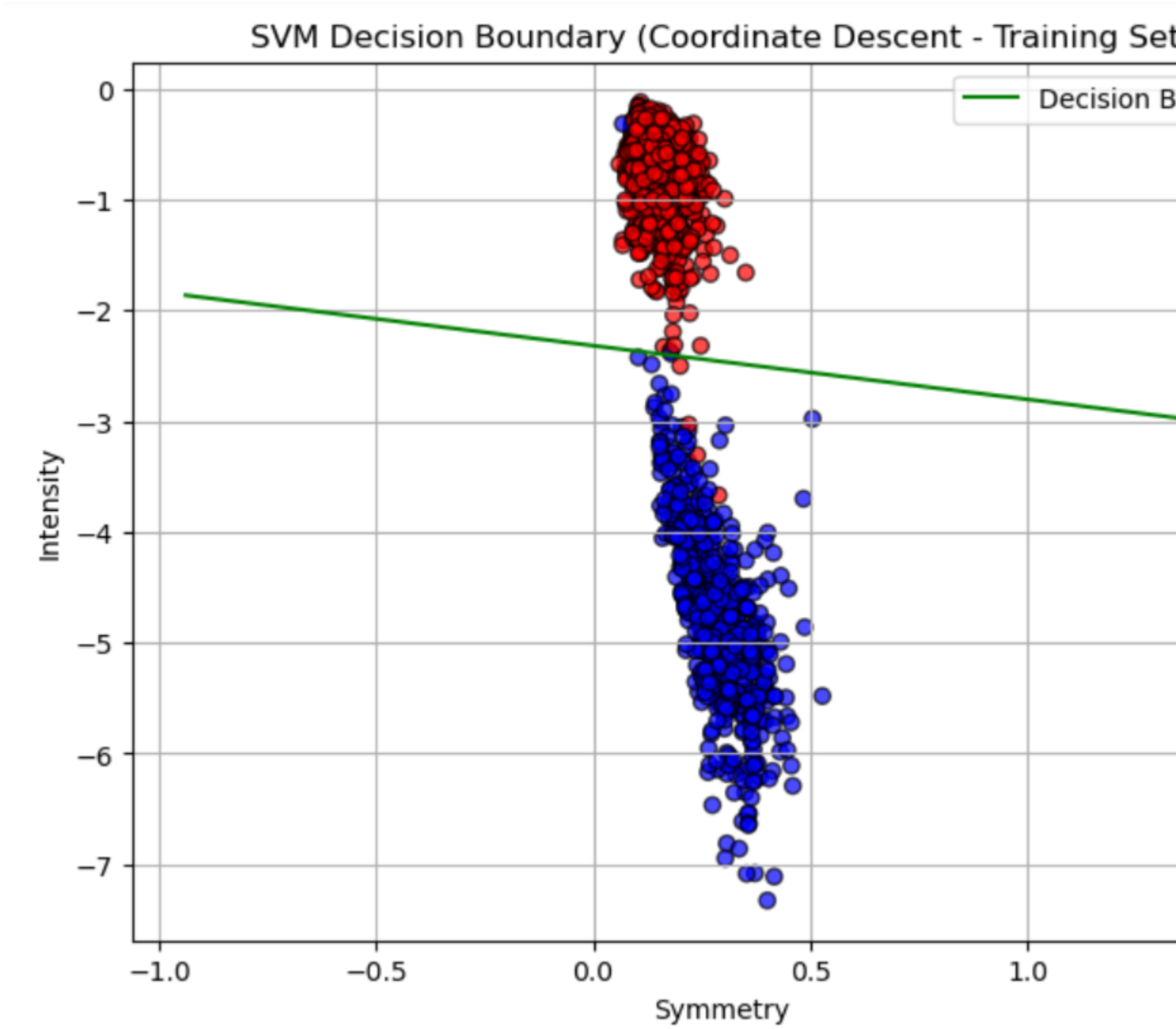
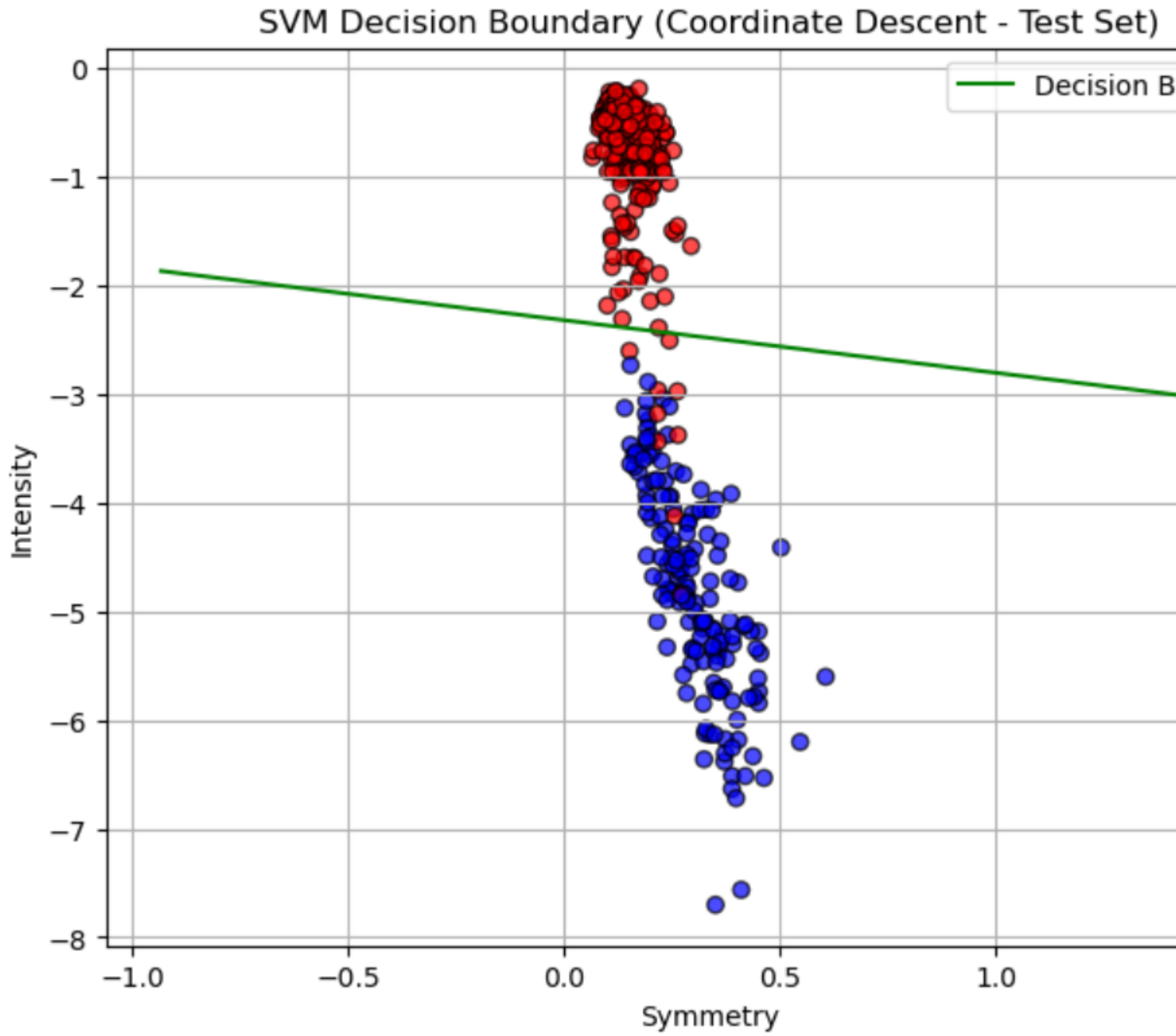

SVM Decision Boundary (Gradient Descent - Training Set
Decision
Intensity
Symmetry
0
−1
−2
−3
−4
−5
−6
−7
−1.0
−0.5
0.0
0.5
1.0



SVM Decision Boundary (Gradient Descent - Test Set)
Decision
Intensity
Symmetry
0
−1
−2
−3
−4
−5
−6
−7
−8
−1.0
−0.5
0.0
0.5
1.0


SVM Decision Boundary (Coordinate Descent - Training Set
Decision B
Intensity
Symmetry
0
−1
−2
−3
−4
−5
−6
−7
−1.0
−0.5
0.0
0.5
1.0


SVM Decision Boundary (Coordinate Descent - Test Set)
Decision B
Intensity
Symmetry
0
−1
−2
−3
−4
−5
−6
−7
−8
−1.0
−0.5
0.0
0.5
1.0